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Model Answer



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[Answer](#)

1. a. Since we replace the first card, the two draws are independent and so $P(HH) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$, and

$$P(\text{Diff}) = P(H\bar{H}) + P(C\bar{C}) + P(D\bar{D}) + P(S\bar{S}) = 4 \left(\frac{1}{4} \times \frac{3}{4} \right) = \frac{3}{4}$$

- b. Since we now do not replace the first card, the two draws are not independent and we must use the conditional probability:

$$P(HH) = P(H_2|H_1)P(H_1) = \frac{12}{51} \times \frac{1}{4} = \frac{3}{51}, \text{ and}$$

$$P(\text{Diff}) = P(\bar{H}_2|H_1)P(H_1) + P(\bar{S}_2|S_1)P(S_1) + P(\bar{D}_2|D_1)P(D_1) + P(\bar{S}_2|S_1)P(S_1)$$

$$= 4 \left(\frac{39}{51} \times \frac{1}{4} \right) = \frac{39}{51}$$

2. $P(F|S) = \frac{P(S|F)P(F)}{P(S)} = \frac{0.9 \times 0.005}{0.15} = 0.03$, i.e. in this framework, there is only a 3% chance of a dangerous fire, given smoke.

3. 1. $A \cup (\Omega \setminus A) = \Omega$ and $A \cap (\Omega \setminus A) = \emptyset$, so $P(A \cup \Omega \setminus A) = P(A) + P(\Omega \setminus A) = 1$.

2. $B = A \cup (B \setminus A)$. Since $A \cap (B \setminus A) = \emptyset$, we have $P(B) = P(A) + P(B \setminus A)$.

3. $A \cup B = A \cup (B \setminus A)$ and $A \cap (B \setminus A) = \emptyset$. So:

$$P(A \cup B) = P(A) + P(B \setminus A) = P(A) + P(B \setminus (A \cap B)) = P(A) + P(B) - P(A \cap B)$$

